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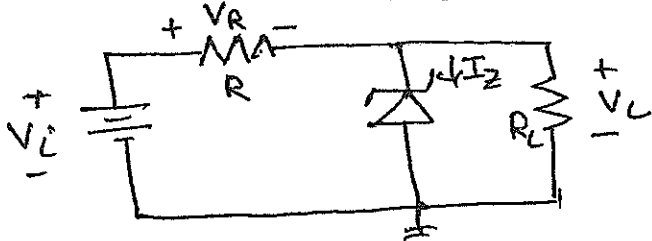
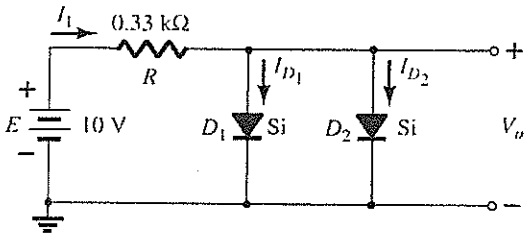
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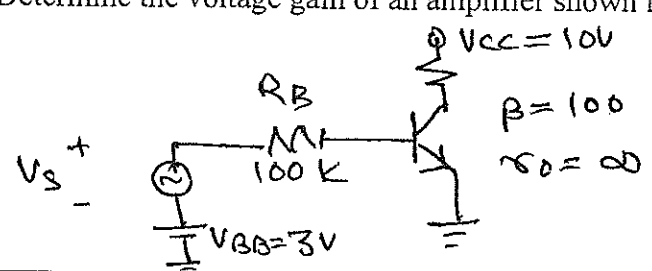
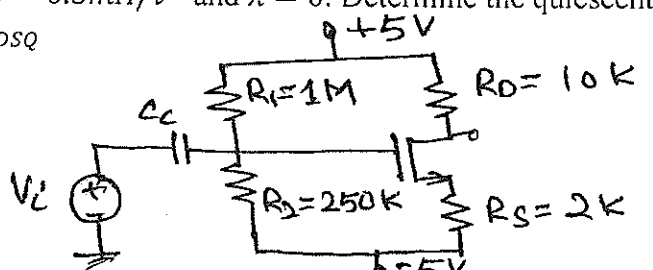
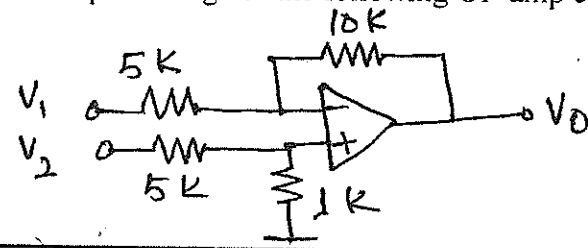
B.Tech Second semester
END-SEMESTER EXAMINATION, APRIL-MAY, 2024

Electronic Devices and Circuits
Course Code: ECECC202 /ICECC202/EIECC202
Duration: 03 Hours

Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Define mobility, conductivity, and diffusion constants and describe the concept of drift and diffusion currents in semiconductor materials.	5	1
1b	Explain reverse saturation current. Find the voltage at which the reverse current in germanium PN junction diode attains a value of 90% of its saturation value at room temperature.	5	1
1c	Find the concentration (densities) of holes and electrons in n-type silicon at 300 K. Given: $n_i = 1.5 \times 10^{10} / \text{cm}^3$ $\mu_n: 1300 \text{ cm}^2/\text{V} - \text{s}$ $\mu_p: 500 \text{ cm}^2/\text{V} - \text{s}$ Conductivity is 300 S/cm	5	1
Q 2	Attempt any 2 parts of the following		
2a	In the voltage regulator circuit shown in Figure (2-a) given below, let $V_i = 6.3 \text{ V}$, $R = 12 \Omega$ and $V_z = 4.8 \text{ V}$. the Zener diode current is to be limited to the range $5 \leq I_z \leq 100 \text{ mA}$. Find range of load current and load resistance. Also find the power rating required for the Zener diode.  Fig-2-a	5	2
2b	Determine V_o , I_1 , I_{D1} , and I_{D2} for the parallel diode ($V_f = 0.7 \text{ V}$) configuration of following figure 2-b.  Fig.2-b	5	2

2c	Draw the circuit of positive and negative clippers and explain their working.	5	2
Q 3 Attempt any 2 parts of the following			
3a	Draw the Eber's Moll model for NPN transistor and using it write the equations for collector and base currents.	5	3
3b	Draw and explain input and output characteristic for common emitter amplifier. Also, clearly indicate various regions of operation on the output characteristics.	5	3
3c	Determine the voltage gain of an amplifier shown in Fig. 3-c.	5	3
 Fig 3-c			
Q 4 Attempt any 2 parts of the following			
4a	Explain the construction and working of the n-channel enhancement type MOSFET. Also, draw the output characteristics indicating various regions of operation.	5	4
4b	The parameters for the circuit shown in Fig. 4-b are $V_{TN} = 0.6V$, $Kn'(W/L) = 0.5mA/v^2$ and $\lambda = 0$. Determine the quiescent values of I_{DQ} and V_{DSQ}	5	4
 Fig-4b			
4c	Draw the circuit of common source amplifier. Using small signal equivalent circuit derive the expressions for voltage gain A_v , Overall voltage gain G_v , input impedance R_{in} and output impedance R_o .	5	4
Q 5 Attempt any 2 parts of the following			
5a	i) What are the ideal characteristics of an Ideal operational amplifier (OP-amp)? ii) Explain the working of OP-Amp based comparator circuit.	5	5
5b	Draw and explain the circuit of an OP-amp based integrator. Derive input-output relationships and plot both input and output waveforms.	5	5
5c	Find the output voltage of the following OP-amp circuit Fig. 5-c.	5	5
 Fig-5-c.			